EXERCISE 1:

(a)The correlations between the lagged return variables and year's return are close to zero. The substantial correlation is between **Lag2**and **Today**. By plotting the data we see that **Lag2** is increasing over time.

(b)

	coef	std err	z	P>|z|
intercept	0.2669	0.086	3.106	0.002
Lag1	-0.0413	0.026	-1.563	0.118
Lag2	0.0584	0.027	2.175	0.030
Lag3	-0.0161	0.027	-0.602	0.547
Lag4	-0.0278	0.026	-1.050	0.294
Lag5	-0.0145	0.026	-0.549	0.583
Volume	-0.0227	0.037	-0.616	0.538

Based on the logistic regression summary table , I can see that **Lag2** statistically significant since their P-value is less than α = 0.05. The remaining lagged predictors and Volume do not show statistical significance.

(c)

```
Confusion matrix:
 Truth      Down   Up
Predicted
Down         54   48
Up          430  557
Accuracy: 56.1065 %
```

The confusion matrix breaks down the prediction errors into both **Type I errors (False Positives) (430 cases)**—where the model predicted an upward market movement (Up) incorrectly—and **Type II errors (False Negatives)(48 cases)**—where it failed to predict an upward movement.

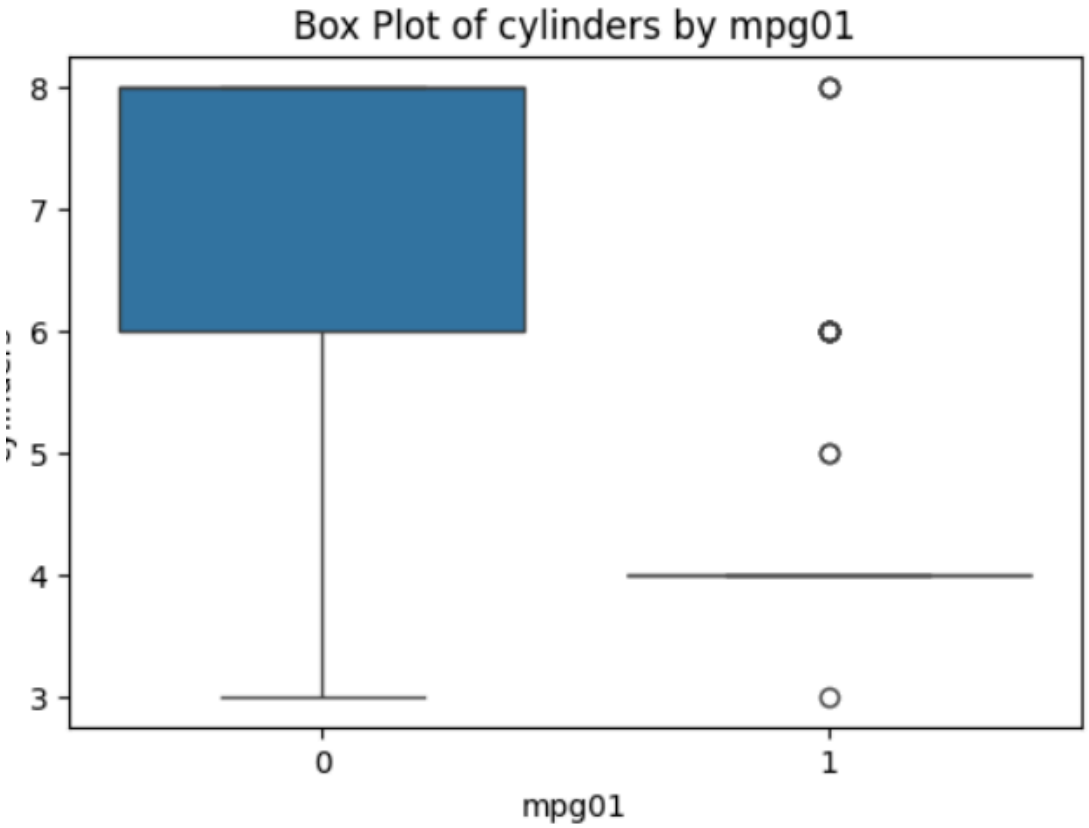
Exercise 2

(b) Logistics Regression and LDA give the best result (both are `0.6250`)

I use combination of **Lag1** and **Lag2** as predictors. Its best method is `Logistic Regression - Accuracy: 0.5769` **`and`** `LDA - Accuracy: 0.5769`

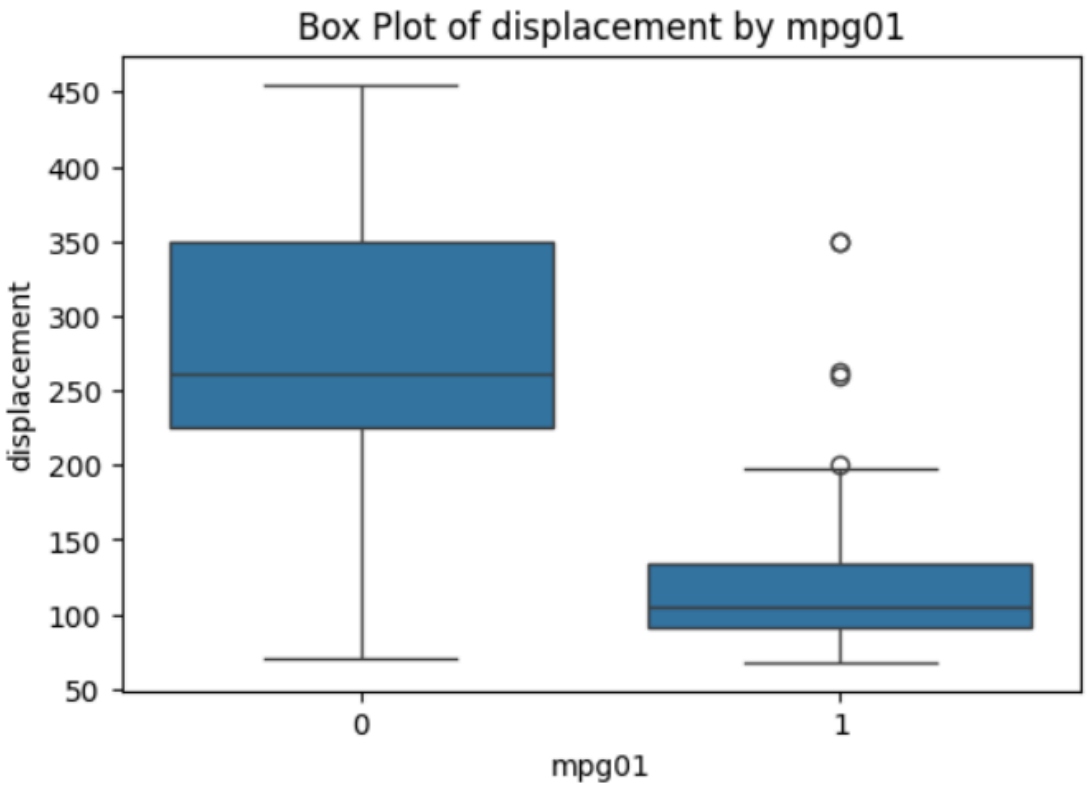
Exercise 3

(b)



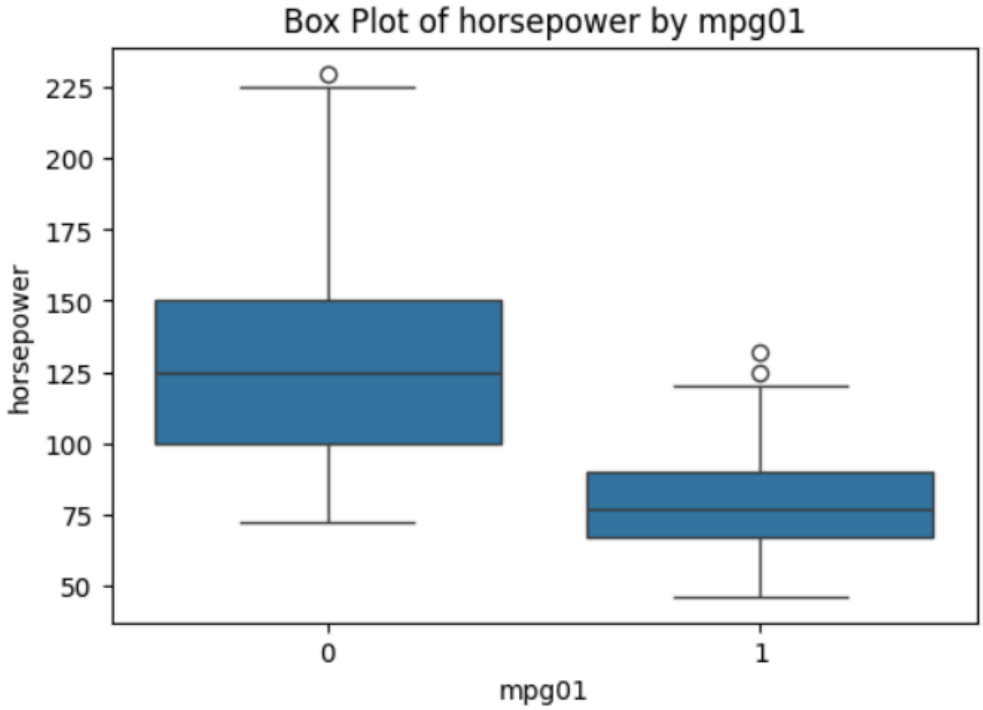
- mpg01 = 0: the majority of values fall within the 6–8 cylinders range.
- mpg01 = 1: the median is 4 cylinders, with outliers present at 3, 5, 6, and 8.
- There is a clear distinction between the two groups; however, the boxplot for group 1 appears compressed into a single line because most of the values are equal to 4.

Conclusion: The number of cylinders is a fairly good predictor of MPG, as vehicles with higher MPG typically have fewer cylinders.



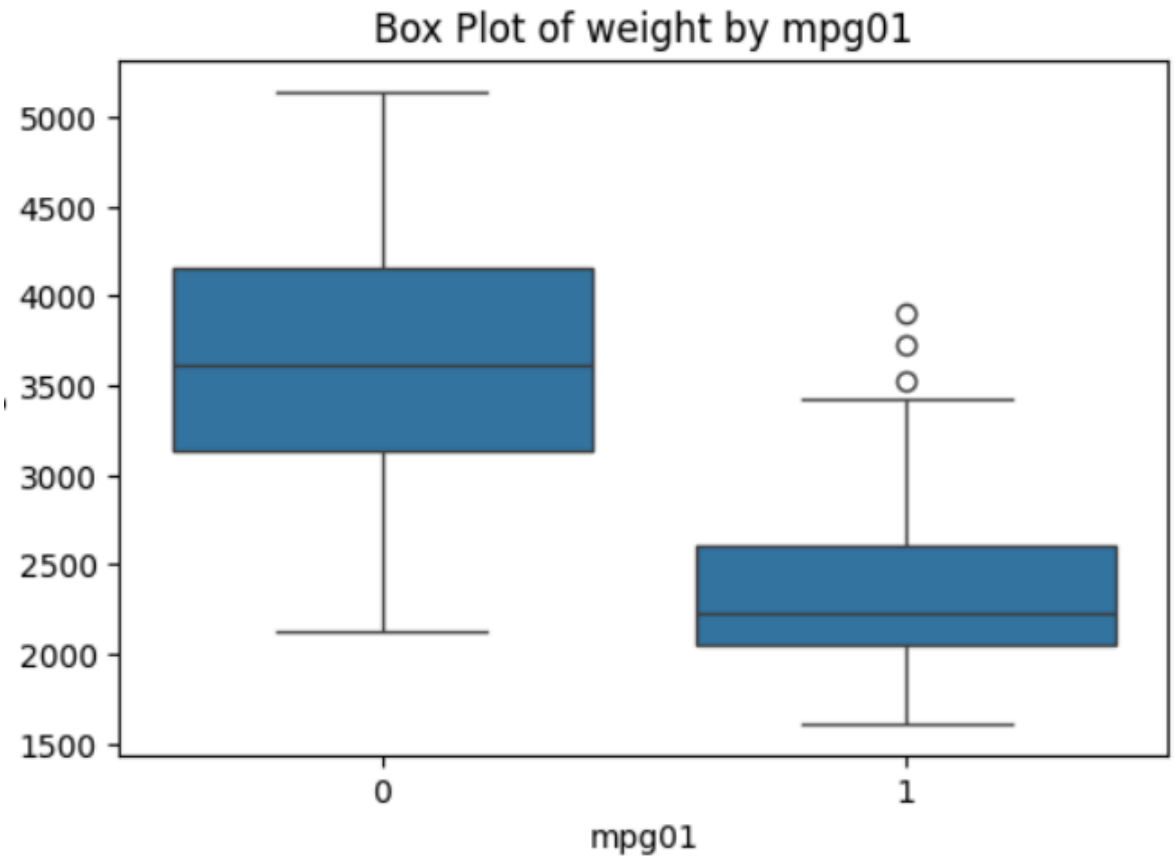
- mpg01 = 0: median is approximately 260.
- mpg01 = 1 group: median is approximately 105.
- The two IQR boxes show almost no overlap, indicating a clear distinction between the two groups.

Conclusion: displacement is a very useful feature for predicting mpg01.



- mpg01 = 0: median is approximately 125
- mpg01 = 1 group: median is approximately 75.
- The two IQR boxes show only a little overlap, indicating a clear distinction between the two groups.

Conclusion: displacement is a very useful feature for predicting mpg01.



- mpg01 = 0: median is approximately 3600.
- mpg01 = 1 group: median is approximately 220.
- The two IQR boxes show almost no overlap, indicating a clear distinction between the two groups.

Conclusion: displacement is a very useful feature for predicting mpg01.

Most useful features is ‘weight’ and ’displacement’

Looking at the boxplots for 'weight' and 'displacement,' the interquartile ranges (from Q1 to Q3—the blue boxes) for group 0 and group 1 are entirely separate, with no overlap whatsoever. This indicates that more than 50% of the data for the high-fuel-consumption group (0) lies completely above the corresponding 50% of the data for the fuel-efficient group (1).

Because the boxes are well-separated, classification algorithms (such as Logistic Regression, LDA, or KNN) can easily establish a threshold value to effectively partition the two groups with a very low risk of misclassification.

(d)

Method	Test Error
Logistic Regression	0.152542
LDA	0.127119
QDA	0.135593
Naive Bayes	0.144068
KNN (K=1)	0.127119
KNN (K=3)	0.118644
KNN (K=5)	0.144068
KNN (K=10)	0.152542

Exercise 4

(b)

```
Random State (Seed) = 0 | Validation Error = 0.0290 (2.90%)
Random State (Seed) = 1 | Validation Error = 0.0250 (2.50%)
Random State (Seed) = 2 | Validation Error = 0.0248 (2.48%)

Validation Error (with student, Seed = 0) = 0.0292 (2.92%)
```

The validation error is relatively stable across different random seeds. It ranges from 2.48% to 2.90%, a difference of 0.42 percentage points. With the student's result, the validation error is 2.92%, which is very close to the previous results. Therefore, the estimate has low variability and the model performs consistently across different data splits.

(c)

```
Validation Error (with student, Seed = 0) = 0.0292 (2.92%)
```

Add the dummy variable `student`. It does not reduce the test error

(d)

```
Model Coefficients (with student):
intercept      -10.495488
balance          0.005709
income          -0.000009
student[Yes]    -0.891798
dtype: float64
```

Students can have a lower default probability at a fixed balance because the coefficient of `student[Yes]` is negative (-0.8918). However, students may have a higher overall default rate because they tend to have different balance distributions. Students may have higher balances, which increase the probability of default. This is an example of Simpson's paradox: the relationship within groups can differ from the overall relationship.

Exercise 5

(b)`=== Logistic Regression ===`

```
Test Accuracy: 0.9357
Confusion matrix:
 Truth        0   1
Predicted
0          104   8
1            3  56
=== LDA ===
Test Accuracy: 0.9240
Confusion matrix:
 Truth        0   1
Predicted
0          105  11
1            2  53
```

Exercise 6

```
exp(coefficient):
 workingday                     1.014773
temp                           2.193047
weathersit[cloudy/misty]       0.927529
weathersit[heavy rain/snow]    0.396022
weathersit[light rain/snow]    0.562255
dtype: float64

Pred Linear Model: 318.44
Pred Poisson Model: 366.59

Number of negative fitted values in M2_lm: 833
```

833 fitted values of the linear model M2_lm are negative.

This is possible because linear regression does not constrain its predictions to be positive. In contrast, the Poisson model uses an exponential link function, so all fitted values are strictly positive. Therefore, negative fitted values are impossible in the Poisson model, making it more suitable for modeling count data such as bike rentals.